Shift Changes, Updates, and the On-Call Architecture in Space Shuttle Mission Control

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Abstract

In domains such as nuclear power, industrial process control, and space shuttle mission control, there is increased interest in reducing personnel during nominal operations. An essential element in maintaining safe operations in high risk environments with this 'on-call' organizational architecture is to understand how to bring called-in practitioners up to speed quickly during escalating situations. Targeted field observations were conducted to investigate what it means to update a supervisory controller on the status of a continuous, anomaly-driven process in a complex, distributed environment. Sixteen shift changes, or handovers, at the NASA Johnson Space Center were observed during the STS-76 Space Shuttle mission. The findings from this observational study highlight the importance of prior knowledge in the updates and demonstrate how missing updates can leave flight controllers vulnerable to being unprepared. Implications for mitigating risk in the transition to 'on-call' architectures are discussed.

Key Words: anomaly, common ground, decision, ethnography, event, knowledge, mutual awareness, observation, plan, shift change, update

'ON-CALL' ARCHITECTURE IN SUPERVISORY CONTROL

In supervisory control domains such as nuclear power, industrial process control, and space shuttle mission control, there has been a widespread trend of reducing large deployments of human personnel continuously monitoring dedicated subsets of process data to minimizing human personnel until a problem arises, at which time additional resources are called in. This 'on-call' architecture has the potential to reduce operational expenses by using the full reservoir of resources only when needed.

For example, during the STS-75 Space Shuttle mission, a tethered scientific satellite unexpectedly separated from the shuttle. As a result, two flight controllers were immediately called in to support the nominally staffed controller responsible for the mechanical systems on the shuttle. The first controller took over the standard operations for the nominally staffed controller. This substitution allowed the nominally staffed flight controller to work with the second called-in controller on developing a way to prevent the astronauts from being electrically shocked when recapturing the satellite.

By definition, with the on-call architecture, personnel are brought in only when a situation is unusual, has begun to deteriorate, or involves high stakes. They are called in as part of an escalation of cognitive and coordinative activities (Woods and Patterson, 2001). There is inherently a 'workload double-bind' (Woods et al., 1994b) in that when the on-call practitioner is most needed to provide additional resources and expertise, the staffed practitioner has the least time to update the incoming practitioner and to coordinate to redistribute the workload.

Therefore, in order to ensure that the on-call architecture functions effectively, we need to identify ways to quickly bring incoming practitioners 'up to speed' without tying up the resources of the staffed practitioners during critical periods. The goal of this research was to better understand how practitioners are currently brought 'up to speed' in a complex, dynamic supervisory control setting. To this end, targeted field observations were conducted of updates in space shuttle mission control: shift change handovers between mechanical flight controllers during the STS-76 mission.

The goal of the shift change handover is to prevent a break in the flow of the monitored process and activities conducted by the flight controllers when there is a change in personnel (e.g., Grusenmeyer, 1995, shift changes in paper mills). A successful handover is defined by a smooth continuity of operations from one shift to the next. There are two senses to this definition. The first is to avoid a rift in terms of interactions with others and ongoing activities being conducted. In other words, the work should continue as if the operator had never been replaced. The second is for the incoming operator to understand what had happened as if he or she had been present and personally engaged in all the activities. The handover update is given to avoid having an incoming practitioner:

- have an incorrect or incomplete model of the process state,
- be unaware of significant data or events,
- be unprepared to deal with impacts from previous events,
- fail to anticipate events,
- lack knowledge that is necessary to perform relevant tasks,
- drop or rework activities that are in progress or that the team has agreed to do, or
- create an unwarranted shift in goals, decisions, priorities, or plans.

The paper is organized as follows. We introduce the domain of space shuttle mission control, including how responsibility is hierarchically distributed and handovers are nominally conducted. We describe how the observational data was analyzed and provide an overview of the observed STS-76 mission. The study findings are then described. Implications of these findings for mitigating risk in two on-call scenarios are discussed.

OVERVIEW OF SPACE SHUTTLE MISSION CONTROL

Hierarchical distribution of responsibility in mission control

Ground-based mission control for the Shuttle Program at the NASA Johnson Space Center (JSC) is responsible for supporting the crew in meeting the objectives of the mission and for ensuring the health of the spacecraft while in flight. The Flight Director (Flight) acts as the central decision maker and coordinates the information flow between the various flight controllers responsible for subsystems of the shuttle. Flight and the approximately sixteen main controllers sit in assigned positions in the 'front room'. Various support personnel, known as the 'back room controllers,' support the front room controllers. For example, for the Maintenance Mechanical Arm and Crew Systems (MMACS) team, the front room position is called MMACS and the back room controllers are Mechanical (Mech I and II), In-Flight Maintenance (IFM), Photo / TV, and Escape. The observations were conducted at the back room Mech I console position. The Mech is responsible for the health and safety of mechanical systems such as power units, heaters, and payload bay doors.

The MMACS team is responsible for ensuring the health and safety of the orbiter's structural and mechanical subsystems during a mission. Flight controllers do much more than continuously monitor system parameters for anomalous readings. Although this is a critical task, there are many subtleties and complexities in the functions that they fulfill, particularly since surprising events are common during space missions. Controllers must exhibit creativity, the ability to work with others, and deep knowledge not only of their mechanical systems but also of the rationales and risk trade-offs behind the flight rules so that they can be applied or modified for the specific circumstances.

Handover updates

Three shifts a day are scheduled when the shuttle is in orbit. Each shift change, or handover, is scheduled for one hour.

Handover updates are designed to have information flow bottom-up through the hierarchy of the incoming shift. For every position, the outgoing flight controller updates the incoming controller, both physically co-located at their assigned consoles. These primary briefings are essentially private (i.e., without using the voice loop communication system described in Patterson, Watts-Perotti, and Woods, 1999), with the convention that no one is allowed to interrupt these communications. After the intensity of the primary briefings has died down, the incoming back room controllers (e.g., Mech) brief the incoming front room controllers (e.g., MMACS). These briefings are used to check the understanding from the primary briefings and coordinate the activities to be conducted during the shift. This update is conducted on a dedicated voice loop channel (e.g., MMACS ALT) on which the flight controllers speak using headphones with audio hookups so that other controllers can listen in on the communications. In parallel with the briefings from the back room to the front room controllers, the incoming front room controllers give the incoming Flight Director a short, high-level update on a voice loop that is dedicated for this purpose (AFD CONF). These briefings are closely monitored by the entire mission control center, which serves to check the shared understanding of the situation following the various discipline handovers.

METHODS

Sixteen of twenty-seven handovers in the mission control center (MCC) at the NASA Johnson Space Center were directly observed during the Space Shuttle mission STS-76 [EP]. The 16 observed handovers were divided between the three shift transitions (5, 6, and 5). The naturally occurring verbal behavior was audio-taped. In order to minimize the effect of observation on the flight controllers' behavior, previous observations had been conducted with the controllers and questions to clarify the content of the handovers were asked only after the handover was completed.

The raw data included field notes of face-to-face and voice loop verbal communications and copies of flight documentation such as handwritten logs and flight plans. The data was analyzed iteratively, using theoretical frameworks to recognize and abstract relevant patterns (Hollnagel et al., 1981). Process tracing protocols (Woods, 1993) for each handover were created that described the activities in domain-independent terms and separated the communications made by the different participants (Figure 1). One-page summaries for each handover were generated and patterns across the handovers related to the research question were identified.

As has been noted by many ethnographic, or "cognition in the wild", researchers, observation and analysis is heavily influenced by the theoretical frameworks that are used to recognize and abstract patterns in complex data. Three frameworks in particular guided the observation and data analysis in this study: 1) dynamic fault management, 2) distributed replanning in anomaly response, and 3) common ground in communication.

The first conceptual framing of the flight controller's task was dynamic fault management (Woods, 1994a). With this framing, a controller recognizes unexpected findings in the data stream, conducts diagnostic searches, and generates hypotheses about faults that could account for the observed pattern of disturbances. This reasoning process goes on in parallel with interventions intended to either protect systems, i.e., "safing" interventions, or to gather additional information, i.e., diagnostic interventions. For a difficult anomaly, there can be challenges in diagnosing the anomaly, figuring out its impacts on related subsystems, performing safing activities in parallel with troubleshooting, and deciding whether or not to obtain more data. Based on this framework, during the transfer of responsibility in a shift change, updates were anticipated to potentially include: 1) unexpected findings in the data stream that might be symptoms of system faults, 2) diagnostic hypotheses to account for unexpected findings, 3) impacts of faults on the monitored systems and other agents, 4) cascading events that were triggered by a system fault, 5) diagnostic interventions, and 6) safing interventions.

Second, it was believed that several important elements might not be covered by the dynamic fault management framework alone that are important in distributed, dynamic, event-driven supervisory control. First, although theoretically dynamic fault management could be accomplished by a team as well as an individual, the implicit assumption of the framework is that planning can be conducted by the agent or agents performing dynamic fault management largely independently of the goals and plans of other agents. In other words, there are few interdependencies with agents external to the immediate team to take into account with respect to dynamic fault management activities. In space shuttle mission control, the coordination required by agents is very complex. Distributed replanning is a critical component of anomaly response (Woods, 1994a). In distributed replanning, multiple people supported by computerized systems assess the implications of an unexpected finding, or anomaly, for planned future activities, evaluate contingencies, and modify plans in progress. During replanning, coordination across multiple people in different roles is more complex than assigning and synchronizing tasks. As part of this coordination, teams of people adopt and portray stances about critical decisions that affect multiple agents. The concept of stance is a combination of a position towards a significant issue (i.e., a decision a team faces) and the rationale for that position, which is often predictable given the position on a tradeoff function associated with particular roles. For example, mechanical systems controllers might be more concerned with determining the cause of a malfunction than controllers primarily tasked with the safety of the astronauts, such as the flight surgeon. Based on this framework, during the transfer of responsibility in a shift change, updates were anticipated to potentially include evidence of discussions about: 1) plans, 2) stances, 3) goals, 4) positions on tradeoff functions, 5) contingencies, 6) intent, 7) impacts to previously planned activities and expectations within the team, and 8) impacts to previously planned and expectations of other teams.

Third, the goal of the observed updates could be framed as creating and maintaining a common understanding, or common ground (Clark and Brennan, 1991) between human agents. This common ground is what would allow the practitioners to accept the responsibility and authority associated with a position for a period of time without being taken by surprise. As others have observed, the notion of common ground is a complex conglomerate of many interdependent elements, including the interacting elements of: 1) knowledge that is known to be shared between individuals (Clark and Brennan, 1991, Wegner et al., 1985, Hutchins, 1995), 2) shared goals or intentions, 3) mutual beliefs about the current state of affairs and the predicted effects of actions on the state of affairs (Clark, 1992, Suchman, 1987, Clark and Brennan, 1991), 4) shared awareness of others' activities and the state of the monitored process, and 5) common frames of reference (e.g., fixed line diagram in London Underground line control room, Heath and Luff, 2000). The conceptual framework of common ground influenced the data observation and analysis in that updates to relatively ungrounded controllers, such as the update immediately following ascent, were anticipated to have a different character than updates based upon a more established common ground. In addition, deviations from expectations, including unexpected data and changes to the plan, were expected to be highlighted more than data and plans that conformed to prior expectations. Finally, it was anticipated that controllers might explicitly use strategies that built upon existing shared understandings in the updates, such as by implicitly assuming that some topics and sub-topics would not need to be included and using coded language to communicate more efficiently.

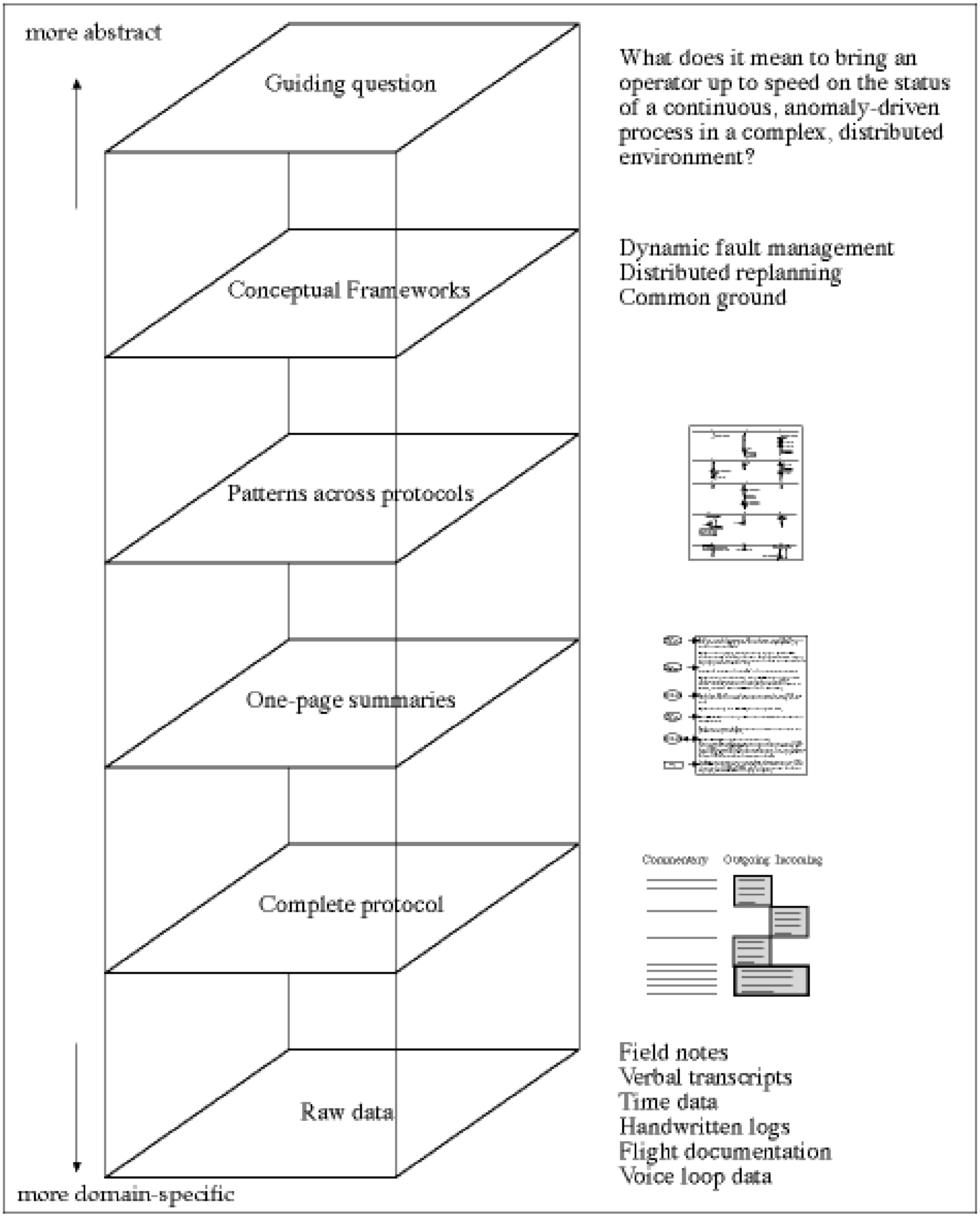

Figure 1. Using conceptual frameworks to guide data abstraction and analysis

THE OBSERVED STS-76 MISSION

The STS-76 mission included a rendezvous docking with the MIR Space Station. As a result, there was a very short liftoff window (seven minutes instead of several hours) and the MMACS team had to monitor specialized docking mechanical systems. Due to the additional workload, the back room Mech position was staffed for the entire flight instead of only during the high-tempo periods such as ascent and entry, which is the staffing configuration for nominal missions.

The initially scheduled liftoff was postponed for one day because of high winds and rough seas at Cape Kennedy (Figure 2 provides an overview of the mission events). The second liftoff attempt began without incident at 2:13 a.m. on March 22, 1996. During ascent, two anomalies in the systems under the responsibility of the MMACS team were observed: a freeze in the Water Spray Boiler (WSB) that cools the third Auxiliary Power Unit (APU), and a hydraulic leak on the third APU.

Both anomalies were definitively diagnosed and neither was severe enough to require an aborted ascent, so the shuttle attained its planned orbit altitude and most of the ascent mechanical systems were shut down. The first anomaly, the Water Spray Boiler (WSB) freeze-up, is a relatively common problem with a well-defined response procedure that mainly involves verifying that the WSB works a day before entry. Although this procedure could not be implemented because the water spray boiler was on the same system that had the hydraulic leak, the WSB freeze-up did not cause an escalation of cognitive and coordinative activities because the procedural action was not required for several days and the eventual decision to assume that the WSB would be operational for entry without the standard test was not contested.

The second anomaly was significant and novel enough to create an escalation of cognitive and coordinative activities. The MMACS controller with specialized knowledge of the Auxiliary Power Unit (APU) immediately called himself in, based on watching the ascent on NASA Select TV, to provide expertise in deciding whether to shorten the mission duration. The decision was made not to shorten the mission because the leak was small enough that some capability remained in the APU system and the leak was unlikely to get much worse during the generally quiescent orbit configuration. There were cascading repercussions from this anomaly to several other aspects of the mission as actions were taken to protect the leaking hydraulic system, both to maintain effective redundancy of critical systems, and to protect the MIR Space Station from contamination (Table 1). Several of these planned potential actions were debated by additional called-in operational, engineering, and management personnel to ensure that the plans were robust to contingencies.

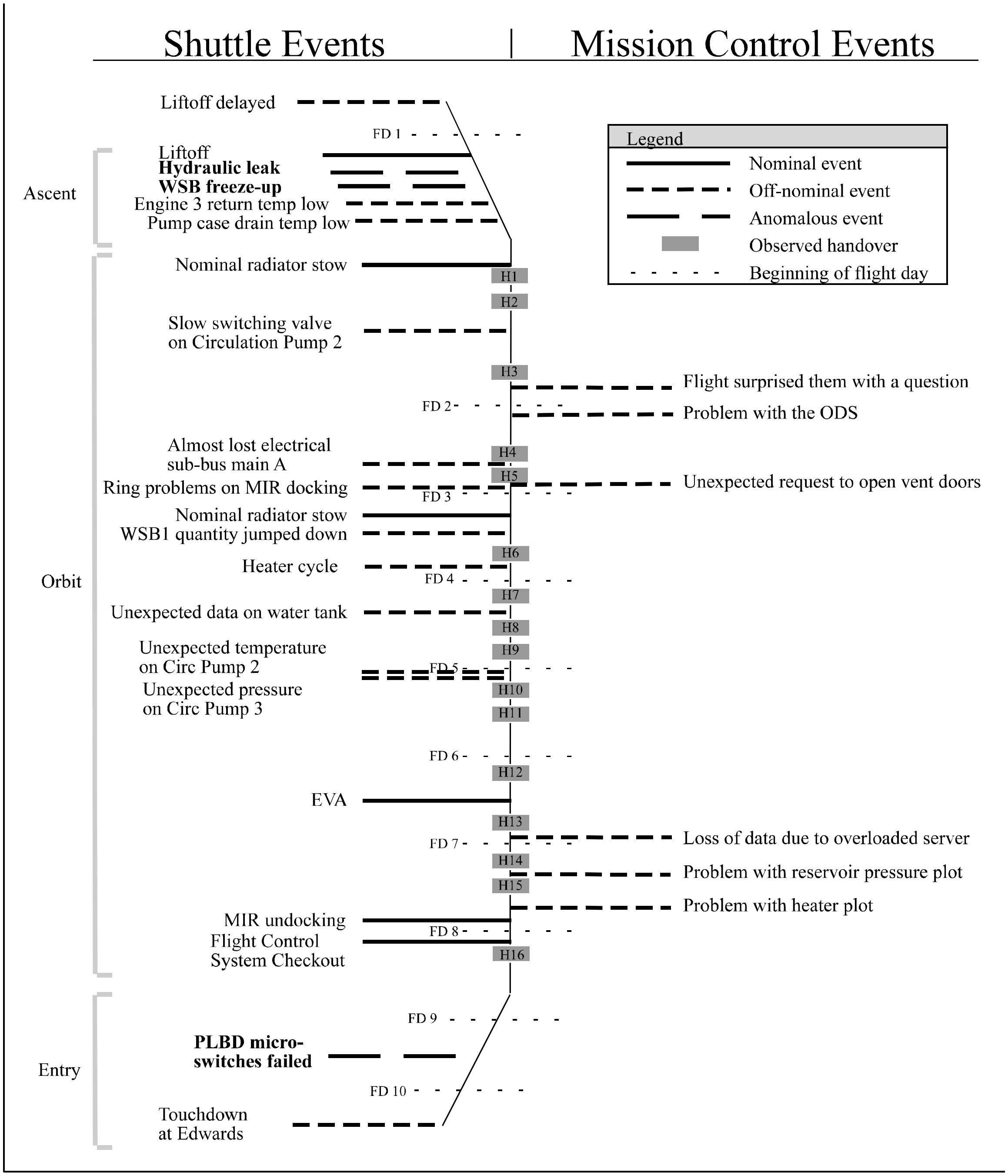

Figure 2. Overview of events and observed handovers in STS-76 mission

TABLE 1

Changes to Plans as a Result of the Hydraulic Leak

Changes to plans	Rationale
Minimize circulation pump operations	To minimize the use of the leaking APU
Close vent doors before docking with the station	To protect the space station from hydraulic fluid
Use a circulation pump instead of an APU to check the flight control system	To reduce the risk of losing redundancy on APUs
Use 2/3 APUs for entry	To avoid relying on the leaking APU
Land at Edwards not Kennedy Space Center	To minimize stress from crosswinds on APUs

On flight day 8, the decision was made to come home one day early (flight day 9 instead of flight day 10) due to weather predictions at Kennedy Space Center (KSC) and concern about the reduced redundancy in the APUs due to the hydraulic leak. On flight day 9, however, both opportunities for entry were waived off because of fog and unpredictable weather at KSC. The astronauts prepared to spend another day in orbit, expecting to land on flight day 10.

When the decision was made to stay on orbit for another day, the payload bay doors were commanded to open but the procedure automatically halted when the sensor indicated that one latch was still closed. After the crew visually determined the latches to be in an open configuration, it was assumed that the sensor was giving an erroneous indication, and the doors were commanded open manually. They opened without further incident. If the payload bay doors had not opened, the shuttle would have had to make an immediate emergency landing.

On flight day 10, the second landing opportunity at Edwards Air Force Base was taken after waiving off the first landing at KSC due to poor weather conditions. The decision to land at the less preferred Edwards site, which requires expensive ground transport back to KSC, was made in order to have better weather conditions, particularly lower crosswinds. The shuttle therefore touched down at Edwards at 7:29 a.m. March 31, 1996, and responsibility for the orbiter transferred from the flight controllers at the NASA Johnson Space Center to other NASA organizations.

FINDINGS FROM THE OBSERVATIONAL STUDY

It might be expected that, during the hour scheduled for each handover, the incoming controller would immediately and continuously receive verbal updates until the outgoing controller departed. This situation was not the case in any of the observed 16 handovers. In every handover except the handover immediately following ascent (which had been personally observed by the incoming controller) and Handover 9 when the incoming controller read a packet of information left by the outgoing controller, the controllers engaged in short high-tempo briefings about 20 minutes after the incoming controller arrived (Table 2). During the time prior to the update, the incoming controller would generally sit next to the outgoing controller while listening to the voice loops, monitor the data screens, and look through the flight log and other documentation. One controller (personal communication) described his opinion about the reason why handover updates often do not begin immediately upon arrival of the incoming controller:

> "You can see during handover that one of the first items that would happen is that the oncoming shift, the incoming shift, would sit down and read the previous two shifts since he was in. And see what had happened over the 16 hours since he had been in. They would sit down and discuss it with the person that they're taking over from and any other little innuendos that haven't been mentioned in the log so that they are well aware that everything that has happened up until that point in time. Because when that person goes home, you know, they don't have any insight. So if there's anything further coming up…then they're not surprised by it, they know about it and they're well aware of it.

They know who else is aware of it…It's a good system. We couldn't operate without logs…Very important stuff."

Also during the handover time, the incoming controllers would occasionally brief their incoming superiors over the voice loops, who would then brief their superior, the incoming flight director. The position responsibility was officially handed over when the incoming controller switched from an alternate to a primary team voice loop and the handover officially ended when the flight director from the outgoing shift verbally released the outgoing controllers via the Flight Director voice loop. In several instances, outgoing controllers stayed beyond the official end of the handover to perform specific activities or attend meetings related to the hydraulic leak anomaly.

TABLE 2
Length and start time of observed handover briefings

Handover	Handover start time	Briefing (min)	Primary briefing start time (min)	Voice Loop briefing start time (min)	Handover Duration (min)
1	4:20	1	20	N/A	40
2	5:56	14	22	36	42
3	16:18	20	32	68	88
4	14:51	10	0	N/A	39
5	20:58	15	15	N/A	55
6	23:41	10	0	33	49
7	1:25	11	28	N/A	39
8	9:36	8	4	39	42
9	17:00	N/A	40	N/A	N/A
10	1:34	1	6	N/A	41
11	7:20	8 (5+3)	15	N/A	137 (includes meeting)
12	9:12	14	0	N/A	14
13	16:52	9 (2+7)	42	N/A	71 (includes meeting)
14	9:45	4 (1+3)	47	0	50 (includes troubleshooting)
15	17:02	13	13	N/A	31
16	11:53	10 (2+8)	27	N/A	40
Avg		9.87	19.44	35.20	43.33
St Dev		5.17	15.53	24.16	17.17

The findings from the observations highlight the influence of prior knowledge on the updates and how missing updates can leave flight controllers vulnerable to being surprised or unprepared. First, the incoming controllers initiated many of the topics in the handover updates, demonstrating shared knowledge about what topics would be important to cover in the handover. Second, incoming controllers were observed to ask questions that were highly specific and indicated a detailed knowledge of the current status of a particular topic item, offloading much of the work necessary by the outgoing controller to determine what the incoming controller needed to learn. Third, the content of the handovers heavily emphasized events and activities, data analyses, and decisions that were triggered by the escalating event of the hydraulic leak anomaly. Finally, although many of the updates were effective in bringing the incoming controllers up to speed, an incident was observed where a controller was surprised by a request to close the vent doors because he had not been updated that there had been a reversal to a prior decision not to close the doors.

Mixed-initiative interactions: topic initiations by incoming and outgoing controllers

Handover updates fluidly shifted from one topic to another. Handover 13 (Figure 3) between the outgoing and incoming back room Mech controllers is used to illustrate how topics were initiated during the handover updates. Above each line is a description of the topic that is introduced by either the outgoing flight controller (on the left) or the incoming flight controller (on the right) and below the line is the beginning of the dialogue on that topic. The entire briefing took nine minutes, divided into two segments of 2 and 7 minutes due to a pre-arranged side meeting with another person. The update, like all of the handovers, began with a recognizable signal that the controller was willing to initiate the briefing: "Anything going on?" Following this initial question, the controllers began discussing a meeting between the mission controllers and engineers about impacts to the operational plan due to the hydraulic leak anomaly. Many of the other topics discussed during the handover were continuations of ongoing replanning efforts for entry procedures as a result of the hydraulic leak in the auxiliary power unit, particularly contingency planning for cases such as loss of another auxiliary power unit or high crosswinds. Note that at the end of the update, the incoming controller re-initiated a previous topic, changes to the shutdown procedure for the auxiliary power unit. This is likely because he wanted to engage in a lengthier debate on the topic than would have been appropriate earlier in the briefing.

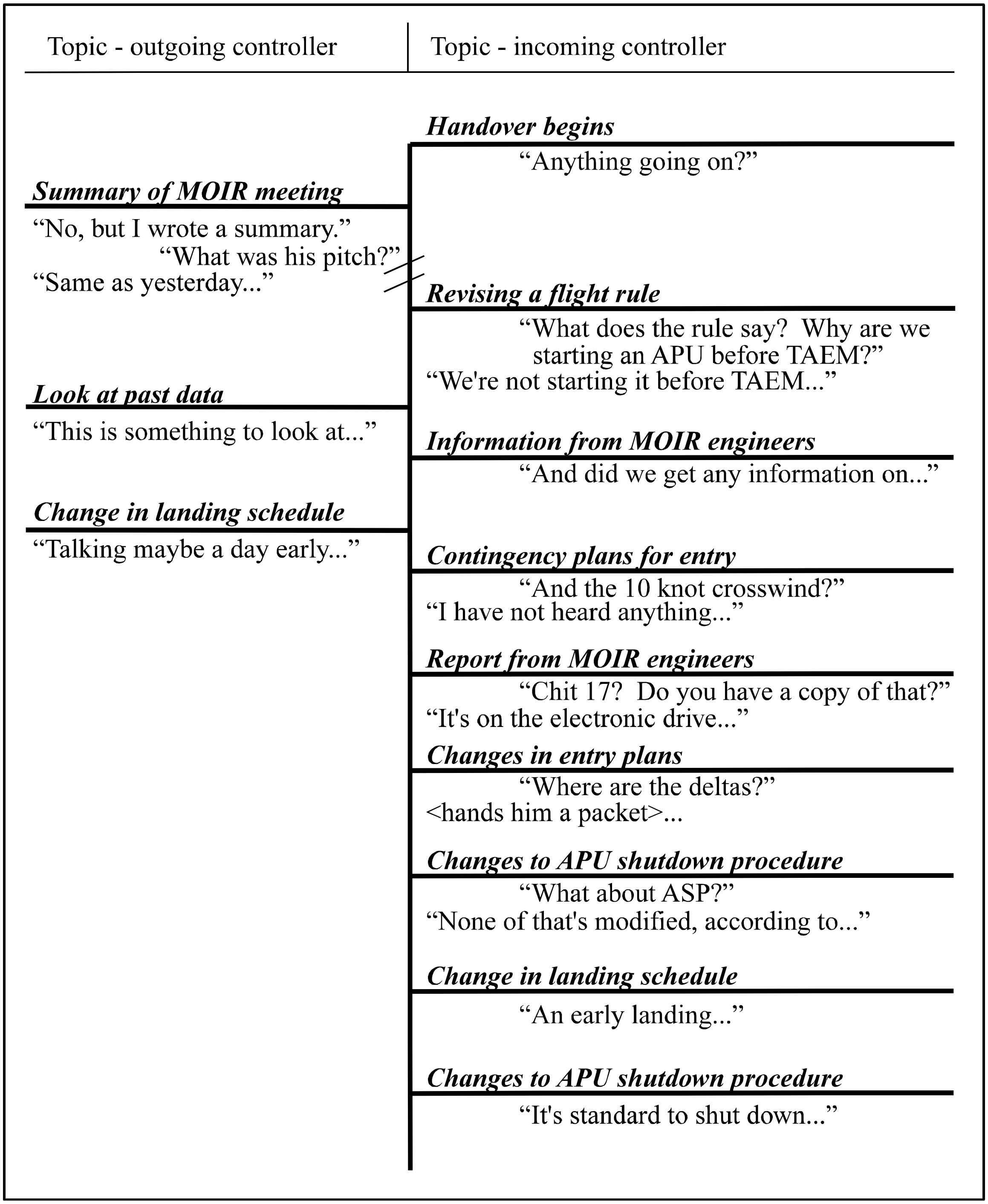

Figure 3. Topic initiations in handover 13

It is a clear pattern across multiple handover updates that topics were initiated by both outgoing and incoming controllers. Since the controller who worked the previous shift should theoretically have more knowledge than the person being updated should, it follows that the expectation would be that the outgoing controller would initiate most of the topics. Nevertheless, it is apparent that incoming controllers initiated many of the topics in the handover updates (Figure 4). At an a α level of 0.01 with the t-distribution, the confidence interval for topics initiated by incoming controllers is [1.5, 7.0], which is clearly greater than zero. In addition, a one-tailed t-test comparison of the number of topics initiated by outgoing and incoming controllers gives a p value of 0.08, which is suggestive but not conclusive that outgoing controllers initiated somewhat more of the topics in the handover update.

The likely explanation for this finding is that incoming controllers had prior expectations about the topics that would be important to discuss before initiating the update. Not only were the incoming flight controllers directly involved in the ongoing activities two shifts before the update occurred, they also had probably read the handwritten log, looked at the events that were being tracked, looked at the flight plan for the day, and listened to the voice loops for some time. Because the incoming controllers had this mission-specific knowledge in addition to their general heuristics about activities in mission control, they could anticipate the important topics to be discussed. Note that the structure of handover 2 supports this explanation. The update in handover 2 was given to a practitioner who was beginning his first shift of the mission. In this handover, the outgoing controller initiated most of the topics. Note that the same personnel were involved in handovers 8 and 14, so the structure of handover 2 was probably not a result of individual personality factors but a function of the incoming controller being less aware of the important topics to cover in Handover 2. Similarly, in handover 16, the topic initiations were dominated by the outgoing controller. This pattern is likely because the incoming controller did not have an up-to-date situation awareness, either because he was substituting for the nominally staffed controller or because it was the last handover before entry and so many of the decisions had been recently finalized.

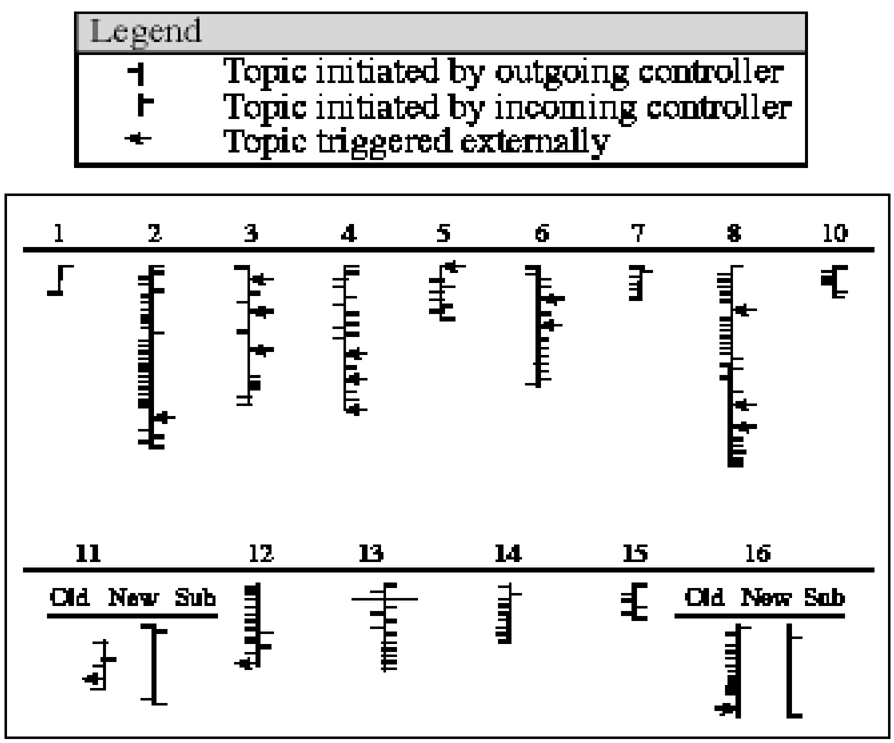

Figure 4. Topic initiations by incoming and outgoing controllers

Questions asked by incoming controllers demonstrated prior knowledge

In addition to analyzing topic initiations, we wanted to characterize the questions asked by incoming controllers during the primary handover briefings. The question categories were iteratively characterized bottom-up from the data mainly with relation to the amount of shared understanding indicated by the question (Table 3). The categories that iteratively emerged from the data analysis were: 1) update initiation questions, 2) topic initiation questions, 3) questions to obtain more details, 4) confirmation questions, and 5) error-checking questions. The questions asked by incoming controllers were used in the handover updates to steer the outgoing controller to specific areas. The most common type of question was where an incoming controller targeted specific information in a topic area about which he or she wanted more details. These types of questions illustrated that the two controllers in the briefing shared much common ground on which to base the update and allowed the incoming controller to narrowly target information which was needed and known to be needed based on the preparatory work of the incoming controller reviewing the documentation and listening to the voice loop discussions.

Although the majority of the questions that were asked were done with the purpose of making the incoming controller more knowledgeable in preparation for transferring responsibility, an additional function of the questions asked during the handover was to perform error checking. In this sense, an additional benefit of the handover was to bring a fresh perspective to the decision making and planning processes, which presumably would increase the robustness of these activities.

TABLE 3
Questions Asked During Primary Handover Briefings

Update Initiation	**Topic Initiation**	**Obtaining More Details**	**Confirmat-ion**	**Error Checking**	**Misc.**
Anything going on?	Do you have a copy of the write-up on the MER meeting?	Was the pressure low when they were doing it?	Rads are deployed?	Why do you have so many limits?	What?
Other than this, is anything else going on?	Will you get me a copy of that ET sep?	Voids?	It's heater DCU 2 not B, right?	Is it only 17? I thought it was 10 or 14.	You didn't prepare anything?
Anything going on?	Are we doing vent doors?	How about circ pump temps?	We're doing an earlier Ei purge?	Why do we need to do that?	You have that?
Anything going on?	Do we need to work any FCS CO changes?	Wiring?	OK, so take out Bravo?	Why wouldn't we put it to norm press?	What?
Anything significant going on?	What's the circ pump 3 status?	How does that fit in [less fluid]?	They already went up, didn't they?	Do you think we ought to have them open that up?	Did you list your number?
Anything else going on?	How did the rad stow go?	Does the crew go to sleep on the orbit shift?		He doesn't want to do that?	
What all is going on?	Where did all this tire data come from?	As long as it's above what, zero degrees?		Do you know that for sure?	
What else?	Who gets copies of this flight note?	What is the MMACS preference?		Does the switch being in low or norm affect the caution and warning?	
	Anything going with the hydraulic leak?	Why did he catch us off guard?			
	The rudder speed brake is getting cold, don't you think?	Oh, you mean they'll start at TAEM?			

	Flight caught us off guard?	How much leaked?			
	Slow start of circ pump 3?	Do you know why in the TMBU we're doing operating limits?			
	What's going on with circ pump 2?	Did you send the TMBU number?			
	The main pump case drain temps?	Did you take the TMBUs to MOIR?			
	Are they asleep yet?	Will the hose jump around?			
	And did we get any information on the?	Did <previous controller> look at the plan?			
	And the 10 knot crosswind?	Because only what?			
	Where are the deltas?	What was his pitch?			
	What about AESP?	What does the rule say?			
		Why are we starting an APU before TAEM?			
		CHIT 17, what does it say?			
		This is it?			
		Just entry?			
		You have any idea where that is?			
		Do we do steam vent heater activation during this?			
		FAO has already sent it up so do we delete this step or have them deactivate it later?			
		Do you know what the MEWS problem is?			
		Hardware and caution warning for what, the reservoir?			
		Do they know if we're going to have it anymore?			
		What is that, a BFS TMBU?			

Update Initiation Questions. Questions that signaled a readiness to receive the handover update such as "Anything going on?" were used to begin the primary briefings. This type of question was the least informed in that the entire burden for structuring the update rested with the outgoing controller. Variations on this question, such as "Anything else going on?" were used within the updating session to remind the outgoing controller to be thorough in covering all of the important topics.

Topic Initiation Questions. Like the initiation questions, this type of question prompted the other controller for information, but it required that the controller knew that a particular topic existed. Many of these questions were triggered by an incoming controller monitoring other information sources, such as by reading the handwritten log (e.g., "Flight caught us off guard?"), looking at the data screens (e.g., "The main pump case drain temps?"), looking at the mission plan, or listening to a voice loop update.

Questions to obtain more details. The purpose of this type of question was to obtain more details about a topic that was being discussed. In the example in Table 4, the incoming controller asked for details that the controller who had actively been engaged in an activity would know. In this case, the incoming controller obtained information that supported a particular hypothesis to explain the anomalous data without requiring the outgoing controller to update him on all of the potentially relevant details relating to the topic.

TABLE 4
Using a question to obtain more details

Commentary	Outgoing Controller	Incoming Controller
A circulation pump did not work as expected. The outgoing controller tells the incoming controller to investigate this potential problem.	"We had a ratty circ pump. The switching valve didn't change for 27 seconds. We need to pull data on this."	
The incoming controller asks a question to obtain more details.		"Was the pressure low when they were doing it?"
The outgoing controller fills in the details requested by the incoming controller.	"Yeah, the circ pump pressure came up about halfway, toggled, went up all the way."	

Confirmation Questions. Confirmation questions were generally "Yes / No" questions in order to verify that the controllers shared the same knowledge or interpretation (e.g., "As long as it's above what, zero degrees?").

Error Checking Questions. During the updates, incoming controllers were observed to question outgoing controllers in an attempt to identify and correct potentially erroneous assumptions. An example of this "Are you sure?" interrogation strategy is provided in Table 5, where the incoming controller questioned whether putting the leaking hydraulic system on the auxiliary power unit (APU) to a "standby" configuration for use in case another APU failed would generate a false alarm. In this case, the outgoing controller stated a high confidence in his assumption that no alarm would be generated, so there was no direct effect on their decision to re-enable hydraulic pressure on the leaking system. In other cases, erroneous assumptions were discovered and changes to plans implemented as a result of this type of question.

TABLE 5
Checking a potentially erroneous assumption

Commentary	Outgoing Controller	Incoming Controller
The outgoing controller updates the incoming controller about a decision to re-enable pressure on a leaking system.	"Heater system 3. We're going to go ahead and re-enable hydraulic pressure on their system."	
The incoming controller asks if re-enabling pressure will cause alarms to be unnecessarily triggered.		"Do you know that for sure? Does the switch being in low or norm affect the caution and warning?"
The outgoing controller declares that no alarms will be triggered.	"No, uh-uh."	

In summary, incoming controllers were observed to ask questions that displayed a range of prior knowledge, from questions that broadly indicated a desire to begin the handover update to questions that were highly specific, targeting a gap in knowledge about details of a particular topic item or verifying that an understanding was accurate. In only one case did a controller defer an answer to a question to a later time in order to more quickly troubleshoot a server crash. If many question deferrals had occurred during the updates, this would have indicated miscalibration on the part of the incoming controllers as to what was important to discuss. By accurately anticipating where they needed to be informed, incoming controllers offloaded much of the work necessary by the outgoing controller to determine what should be included in the update.

These patterns of mixed-initiative interactions and interrogation strategies suggest that the update is less effortful and more robust when many of the topics are mutually known before the briefing. The outgoing controller is less prone to missing an important topic as the incoming controller can help to remind the outgoing controller of the topics to be covered. The incoming controller can aid the outgoing controller in targeting knowledge gaps during the update. Investing in a common understanding during low workload periods in preparation for unexpected problems, either by listening in on others' conversation, observing others' activities, or providing updates that have not been requested, has been observed to be a strategy in many complex, dynamic domains (e.g., anesthesiology, Johannesen et al., 1994; satellite mission control, Jones, 1995; aviation, Kerns et al., 1998; military aviation, Rochlin et al., 1987; emergency call centers, Benchekroun et al., 1995). An implication of these observations is that the on-call architecture might work more effectively if practitioners who are assigned the responsibility to be called in when an unexpected event occurs invest proactively in learning the important topics that would need to be covered in an update *before* the situation escalates.

Updates emphasized cascades from the escalating event

Analysis of the content of the handover updates revealed that the updates mainly emphasized activities, data analyses, and decisions that resulted from the hydraulic leak anomaly (Figure 5). The activities in the handover updates included activities that had been accomplished in the past, that were ongoing and needed to be continued by the incoming shift, and activities that still remained to be done during the next shift or handed over to future shifts. There were also data analysis results that were described during the handovers that provided further information about the extent of the hydraulic leak, performed either within the MMACS team or by engineering personnel. Finally, controllers discussed changes in decisions to nominal and contingency plans for upcoming landmark events. With every update about a decision, there was an associated update about the stance toward the decision. For example, the stance of the MMACS team toward the configuration for entry was that the Auxiliary Power Unit (APU) with the hydraulic leak should be turned off in order to avoid relying upon a potentially faulty system. By including the stance toward a decision in the update, the incoming controller would be positioned to provide and defend a recommendation in the event that the decision was reopened for debate at a later time.

Note that the handover updates mainly emphasized deviations from the initial plan. The handovers were built on top of a shared understanding of the nominal plan. It should be recognized that called-in practitioners might not have this shared understanding to build upon unless they are specifically provided with that information in advance.

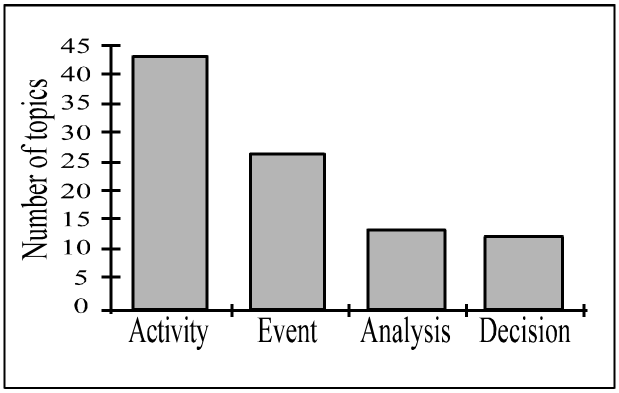

Figure 5. Topics in handover updates

Updates highlighted events

Although many of the activities, data analyses, and decisions discussed in the handover updates were triggered by the hydraulic leak in the Auxiliary Power Unit during ascent, the handover updates also included discussions about events which did not trigger these cascading repercussions. All of the events discussed during the handover updates are shown in Figure 2. The updates included a wide variety of events along a continuum of deviation from expectations: nominal to off-nominal to anomalous to escalating. Generally, the depth of the briefing about the event was a function of how far it deviated from expectations.

Nominal events are defined as events that occurred as planned during the mission. The main events of concern to the mechanical systems controllers (MMACS) that were originally scheduled into the STS-76 flight plan were liftoff, shutdown of the mechanical systems upon obtaining orbit, radiator deployments, radiator stows, extra-vehicular activity (EVA), docking and undocking with the MIR space station, tests of the flight control system a day before entry (FCS checkout), and touchdown. Updates about nominal events were generally brief and mainly given to confirm that an event had occurred as expected (e.g., "They did the EVA."). Not all of the nominal events were mentioned in the updates. In some cases, additional details were provided about exactly what occurred during the event because although it was mostly nominal, there were some aspects that should be noted. In the example in Table 6, the update confirmed that the event occurred nominally and served as a reminder to the incoming controller that only the port radiator was deployed in this case, which was the original plan but normally two radiators are deployed.

TABLE 6
An update about a nominal event

Commentary	Outgoing Controller	Incoming Controller
The incoming controllers asks about the planned event of radiator deployment which was supposed to have occurred in the previous shift.		Rads are deployed?
The outgoing controller implies that the event occurred nominally and reminds the incoming controller that only one of two radiators was deployed, as planned.	Port rad is deployed. We only need the port.	

Off-nominal events are defined as unexpected deviations from the plan that had few impacts to operational plans. The example in Table 7 contains an update about an off-nominal event: a temperature value in the third main engine that was lower than expected. Note that the outgoing controller identified the event based on noticing that the data from one system was lower than data from two identical systems, even though the values were within the hard-coded nominal ranges in the monitoring software. Also, the outgoing controller's update had two related data deviations given sequentially, although he did not explicitly state that the two deviations were related. In addition to giving the observation of the low data, the incoming controller proposed a hypothesis to account for the data, leading fluidly into a diagnostic debate that allowed the two controllers to use each other's expertise to generate and evaluate hypotheses. Finally, the outcome of the diagnostic debate did not include a resolution or selection of a particular hypothesis, even as a working hypothesis, since it was not deemed important to devote the resources to doing so. Had this update been about a large-scale anomaly, selecting and justifying a rationale for an explanatory hypothesis would have been much more important. At this stage, by learning about this deviation in the handover, the incoming controller was prepared to:

- perform the activity of pulling the data,
- alter his expectations for monitoring to track those data points,
- connect that piece of data with other unexplained data, and
- answer questions as they arose.

TABLE 7
An update about an off-nominal event

Commentary	Outgoing Controller	Incoming Controller
The outgoing controller mentions that there is a potential problem with an engine because although the temperature value is in the nominal range, the value is lower than two identical systems.	"System 3 main engine return temp is lower than the other two and I don't know why. So that's a question there."	
He mentions similar data that might also be related because it is on the same system and is also a lower temperature.	"Also, the main pump case drain temp on system 3 was 163 when the other two were 180 post-ascent."	
The incoming controller suggests a hypothesis to account for the unexpected data.		"Hydraulic leak might account for that."
This suggestion triggers an involved diagnostic debate about two possible hypotheses that might account for the data.	(Diagnostic debate about two possible hypotheses, a hydraulic leak and a transducer, that might account for the data.)	

Finally, there were two events during the STS-76 mission that were classified as anomalous in that they were significant enough deviations that they required documented justification of the rationale for diagnosis and response actions taken during the mission, but did not cause an escalation of cognitive and coordinative activities like the hydraulic leak anomaly:

1) a freeze in a water spray boiler that had almost no impact because, based on experience in many past missions where that event was seen and the boiler worked nominally when the coolant warmed up, no immediate action was required, and
2) a microswitch failure on the payload bay doors; had the indication been correct, it would have been a serious anomaly requiring an emergency landing.

It is interesting to note that, although events were clearly critical in the handover updates, the practitioners rarely discussed base data values (e.g., "the pressure is 82 psi"), but rather described data patterns in terms of events that were significant in some way (e.g., "there was a water spray boiler freeze"). In the situation where an automated system would be used to monitor and call in practitioners, it would be important for the system to highlight or visualize significant events, not just plot base data parameters (Thronesbery et al., 1999). It must be recognized, however, that many of the shuttle events, and certainly the associated activities, decisions, and data analyses, would be beyond the capabilities of an automated logger to capture. Therefore, automated monitoring systems would need to be designed such that this other information could be easily annotated by human practitioners at regular intervals in order to avoid called-in practitioners lacking critical information in escalating situations.

The case of the missing update: unprepared to close the vent doors

Although in general the incoming flight controllers took over the responsibility of their positions without incident due in large part to the effectiveness of the updates that they received, an incident was observed where the back room Mech controller did not anticipate a request to close the vent doors prior to docking with the MIR Space Station. The controller was clearly surprised by this request, as evidenced by prior statements made by the controller that he did not believe the action would be requested, a look of surprise when the request was made, and a delay in the timeline because implementing the action took several minutes longer than expected. In addition, the observed controller described the incident to the following shift's controller as: "In the unlikely event that we do it, I didn't want to be stumbling around…then all of a sudden we're doing this…"

The controller was unprepared for the request because he was not updated by another agent in the distributed system about a reversal in the Russian space agency's stance toward the decision about closing the vent doors on the shuttle prior to docking. The inferred evolution of the mindsets of the United States and Russian space agencies regarding whether or not to close the vent doors prior to docking are detailed in Table 8. Normally the vent doors are left open in space to allow oxygen to escape prior to entry. The anomalous hydraulic leak during ascent raised concerns that hydraulic fluid might contaminate the MIR Space Station. Analyses conducted by both space agencies showed that the amount of leaked hydraulic fluid was negligible with the implication that it was not necessary to close the vent doors prior to docking. In addition, NASA planned to conduct a space walk during the mission, demonstrating that they were not concerned about the hydraulic fluid contaminating the interior of the shuttle. During communications between the American and Russian space agencies, the two organizations presented evolving stances toward the decision. One day before docking, the Russians announced that they were "90% go" on docking without closing the vent doors. The observed controller, along with the entire mission control center at NASA Johnson space center, assumed that this was a final decision not to close the vent doors, as evidenced by a voice loop update to the flight director.

Sometime between the conference call and the docking, a representative of the American space agency had a private phone conversation with a representative from the Russian space agency where the decision not to close the vent doors prior to docking was reversed. This decision reversal was never communicated to the personnel in mission control, with the subsequent consequence of the observed controller dedicating his resources to preparing for other tasks and therefore being unprepared for the request.

TABLE 8
Missing update on decision reversal triggers coordination surprise

Time	Inferred Mindset: NASA	Inferred Mindset: Russian Space Agency
0:05 – 2:45	We need to think about how the hydraulic leak will impact the mission.	Same.
2:45 – 8:00	We need to look into whether to close vent doors on the Shuttle to reduce contamination to MIR station.	Same.
8:00 – 17:20	We do not need to close vent doors but we probably will do it to satisfy the Russians.	We should close vent doors to protect the MIR station.
17:20 – 17:23	We are "100% go" on docking. (conference call with Russians)	We are "90% go" on docking. (conference call with NASA)
17:20 – 1:09:00	We will not close vent doors because the Russians did not ask us to.	It is not necessary to close vent doors.
	(PRIVATE CALL)	*reversal*
1:09:00 – 1:17:00	We will not close vent doors because the Russians did not ask us to. **mismatch**	We should close vent doors to be safe as long it will not make things worse.
1:17:57	**SURPRISED** by request to close vent doors.	**SURPRISED** at time taken to close vent doors.

The observation of this instance where a missing update impacted the performance of the staffed controller provides converging evidence that updates are central to effective performance. When practitioners are not fully updated on the current situation, they are vulnerable to these types of 'coordination surprises' (Patterson et al., 1998). Therefore, coping with additional workload during escalating situations by delaying or eliminating updates to called-in practitioners will lead to predictable cognitive and coordinative breakdowns. An essential element in maintaining safe operations with the on-call architecture is to understand how to minimize the effort to bring incoming practitioners quickly and efficiently up to speed in escalating situations.

DISCUSSION

The study findings highlight the importance of updates in preparing incoming practitioners to effectively accept responsibility to be a supervisory controller in a dynamic, event-driven, complex setting and the central role of prior knowledge during the updates. During the update, practitioners learn the status of both the monitored process and distributed agents' activities in response to expected and unexpected changes in the process flow. These observations elucidate why controllers will often refuse to accept a transfer of responsibility from another controller without a face-to-face verbal update. The cognitive impact of the update was observed in all facets of a flight controller's work. The expectations for monitoring were set by knowing what changes had been made to system configurations and what events had occurred. The agenda of activities to be done during the upcoming shift was influenced by knowing what past activities were concluded and what activities were ongoing. Knowing the team's stance toward critical decisions impacted communications with other controllers, particularly when decisions were reopened for debate.

In addition to direct implications for training how to conduct effective shift changes in supervisory control settings, such as by conducting pre-planning for updates by looking at logs and other documentation, these study findings point to other design and organizational implications relating to on-call architectures. Under pressure to be more cost-efficient, NASA and other organizations are interested in using computer-enhanced sensor data processing to enable the reduction of staffing during nominal situations. These findings have implications for two envisioned scenarios where the on-call architecture for supervisory control is used to meet this economic goal. In the first, staffing is minimized until a problem occurs. In this case, the staff must recognize when problems occur and call in practitioners with the appropriate types of expertise to resolve the problem. An example of this scenario is the role of the Station Duty Officer, who is the only staffed flight controller for the US Space Station for all but 3 hours a week, when no crew is onboard the space station. In the second scenario, a computerized system monitors a process and alerts humans when a problem occurs that requires their attention. Although this may seem somewhat futuristic, this scenario is already being considered in several domains, including scientific spacecraft mission control (Brann et al., 1996) and unmanned missions to Mars.

In both of these scenarios, computer-enhanced sensor data processing is required in order to monitor the massive amounts of data in order to recognize significant events that need to be brought to a human's attention. For example, the traditional mission control automated log entries for one console for a few minutes of data is displayed in Table 9 (see Patterson, 1997, for a description of current logging tools in space shuttle mission control). Clearly, if one controller is responsible for monitoring dozens of such consoles at one time, significant events could be missed due to the sheer mass of the data without "intelligent" machine process support to recognize, prioritize, and highlight deviations from expectations.

TABLE 9
Representative entries in traditional mission control automated logs

M23:51:35 MODE SEL MAN ORB UNL V72K2975J has changed from 1 to 0
M23:51:35 MODE SEL MAN EE V72K2976J has changed from 0 to 1
M23:51:55 ORB UNL MODE IND V72X2906J has changed from 1 to 0
M23:54:55 EE MODE AUTO V72K2990J has changed from 0 to 1
M23:51:55 ENTER V72K2982J has changed from 1 to 0

For both scenarios, it is clear from these observations that the event recognition, prioritization, and communication conducted by the mission controllers was much different than that provided in the traditional automated logs. First, the controllers did not communicate base data about "bit flips" on sensor data "changed from 0 to 1." In fact, communications about exact data values was nearly non-existent during the updates. The event descriptions were at a much higher level, such as "ratty circ pump" and "heater cycling" that were based on a complex combination of multiple parameters, not all of which would independently be viewed to be out of normal ranges for most situations and not all of which occurred simultaneously in a discrete fashion. Second, not all of the nominal events were included in the updates, although all of the off-nominal and anomalous events were. Therefore, events that deviated from expectations were treated differently during the updates, and the expectations were highly tailored to what was happening in the mission as well as against a baseline of deviations such as water spray boilers that often freeze up. Third, the event "signature" on which the recognition of the event was based was nearly always more complex than a threshold crossing on a single parameter. For example, one of the events was about a temperature on an engine that was lower than two other engine temperatures. This temperature value was not out of range of nominal parameter values. In addition, there are situations where one would expect the temperature to be lower than the observed value, such as upon entry in the cold atmosphere, which would not constitute an event. Fourth, most automated logging tools only capture and display past information, and much of the handover content related to future events, activities, analyses, and decisions, in order to help the incoming controller prepare for and anticipate these things, or else pass them on to the next shift to do so. Finally, many of the events that were discussed in the handover updates were not about the space shuttle, but about deviations in expected activities, decisions, and plans with other agents, such as the reversal to the decision to keep the vent doors open, and so would be nearly impossible for an automated logger to detect at all due to difficulties in designing sensors to detect those higher-level abstractions.

Overall, one implication of these observations is that automated loggers are currently not capable of completely replacing a human monitor in accurately detecting all unexpected data patterns, prioritizing these patterns, displaying them flexibly at multiple levels of detail upon demand, and quickly filling in targeted holes in knowledge upon request. At the same time, these observations point out how heavily the effectiveness and efficiency of the updates relied upon the incoming controllers having substantial knowledge prior to initiating the update. This observation calls into question whether or not updates to called-in practitioners could be conducted as quickly and effectively as these shift change updates. Without as much prior knowledge, as would be the case in a call-in situation, called-in practitioners would likely take much more time and resources before they could effectively aid the staffed practitioner. It is likely that the update would take longer, and updating would be a larger cost to the staffed practitioner at a very busy time than if the incoming practitioner were more knowledgeable. The burden for thoroughly covering all of the topics to be discussed would fall onto the staffed practitioner, and possibly the called-in practitioner would try to raise topics that are less relevant. Rather than being able to target specific gaps in knowledge through directed questions, the staffed practitioner would be forced to cover more information in the update or risk leaving out important information. Finally, the common ground would not have been built up between the staffed and the called-in practitioner, so the communications would be less terse and rely less on a common body of shared knowledge, leaving open more possibilities for miscommunications. Given that these shift change updates were ten minutes on average and that in escalating on-call situations, ten minutes might be prohibitively long to tie up the resources of the staffed controller, it is likely that other means to prime on-call practitioners to receive updates might become important in effectively drawing in the called-in practitioner in the first scenario.

A partial organizational response to this dilemma would be for called-in practitioners to invest in a process understanding *before* any problems occur. NASA Johnson Space Center has already implemented this organizational solution during missions where the staffing is reduced unless a problem occurs. They have made being on-call an official responsibility that requires investment, although less than if all the controllers are continuously staffed. For each mission, two controllers are assigned the responsibility of being on call, one scheduled from midnight to noon and another from noon to midnight. These controllers observe critical phases of the mission, such as ascent. They also stop by the console in mission control to obtain updates, read the log, listen to the voice loops, and watch the monitored data once a day for about 15 minutes. By investing in learning about events that have occurred during low-tempo periods, they are then more prepared to respond in an on-call situation. We are considering how to additionally support this solution by providing 'open' tools remotely like voice loops and data screens for on-call controllers who are physically and temporally removed from the control center so that they can gain a process feel without leaving their offices. It is also possible that the same tool could be used to provide called-in practitioners with a partial understanding to prime them for the update from the staffed practitioner, thereby reducing some of the burden of updating the incoming practitioner at a busy time.

This field study also suggests implications for the second envisioned on-call scenario where humans are removed from the monitoring loop during nominal situations. In this situation, it is likely that machine processing would have to perform some control activities, not just monitor and record deviations from expectations, in order to reduce the number of times a human agent would need to be called in. From the results of this field study, it is clear that such a tool would probably over-control or inaccurately control a complex process on occasion. Therefore, control actions from such systems should be highly constrained and the consequences of over-controlling or inaccurately controlling should be low.

Based on the results of this study, it must be acknowledged that even with the most advanced automated monitors, it would be dangerous to completely remove human personnel from nominal operations in complex, high-risk environments with escalating events. Automated loggers are mainly used to capture and manipulate data at the level of data parameter values, missing much of the information about significant events, and activities and changes to plans that are associated with cascades from escalating events. Rather than completely replacing human supervisory controllers with automated loggers, perhaps we can develop support tools for human practitioners that are only intermittently involved. For example, we can develop 'hybrid' systems, where humans can periodically annotate information that cannot be captured electronically onto automated logs. These records could then be used to prime called-in practitioners for updates during escalating situations.

The pressure to minimize costs during nominal operations is expected to continue to mount in most supervisory control domains. Given the potentially extreme risks associated with failing to effectively integrate in the additional resources necessary to respond to an escalating situation, there is increased interest in finding ways to mitigate those risks. The findings from this field study highlight the influence of prior knowledge and building a common ground between practitioners in having an effective and efficient update. Investing in a common ground before problems occur, by getting updates from staffed practitioners during low workload periods and 'looking in' on data remotely through computer support tools, will allow practitioners to be more effective at seamlessly providing the necessary expertise and additional resources to safely respond to escalating situations.

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